Robot factory worksheet

Build an offline robot factory from ordered instructions. Start with one fixed pickup fixture and one fixed destination. Pair roles: driver edits; observer predicts and checks; swap halfway. Only the instructor enables supervised hardware.

Pair names: _____ Driver first: _____ Observer first: _____

Minutes	Duration	Session
0-20	20	Meet our factory
20-45	25	Ordered instructions
45-65	20	Observations and uncertainty
65-95	30	First Python edits
95-105	10	Break
105-120	15	Teach inspect and replay demonstration
120-165	45	Explore three stations
165-210	45	Build the physical sequence
210-230	20	Improve one failure
230-240	10	Three pair demonstrations

Make a short conveyor story

Put instructions in order and repeat a small section.

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Your pair can trace the order, predict a repeat count and explain one corrected sequence.

Separate the object from its report

Recognise missing and incorrect observations.

Photo input: Student handbook, Separate the object from its report; learner ZIP notebooks/assets/photo-examples.jpg. Human-visible labels; deliberately invented reports, not camera measurements.: _____

Row 1: actual red, report red. Our decision ...: _____

Row 2: actual blue, report None (missing). Our decision ...: _____

Row 3: actual red, report blue. Our decision ...: _____

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You identify one missing and one wrong report and choose pause/inspect for uncertainty.

Change a name and a number

Read and edit a short Python program before sharing the same program representation with blocks.

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Both partners make one edit and explain it; shared commands are validated before execution.

Watch teach inspect return replay

Understand one saved movement (a stroke) and its required start/end arm arrangement (pose), gripper opening (jaw state) and measured references (calibration).

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You can say teach → inspect/validate → return to start → replay, and point to STOP.

Explore three stations

Connect observations, instructions and the robot model.

Arm and twin observation — our prediction:

_____

Arm and twin observation — run and observed evidence:

_____

Conveyor and observation sequencing — our prediction:

_____

Conveyor and observation sequencing — run and observed evidence:

_____

Offline Blockly and Python preparation — our prediction:

_____

Offline Blockly and Python preparation — run and observed evidence:

_____

We changed one thing: _____

Explain the result and one limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Every pair completes all three positions with one recorded prediction and evidence statement per position.

Dispense carry pick and place

Assemble the supervised fixed pickup to fixed destination sequence.

Our overall prediction: _____

We changed one thing: _____

Dispense — observed evidence: _____

Arrival stopped at pickup fixture — observed evidence: _____

Grasp — observed evidence: _____

Lift — observed evidence: _____

Place and release — observed evidence: _____

Return — observed evidence: _____

Explain the result and one limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Your pair explains the full fixed-fixture sequence and records honest per-step evidence. Physical completion requires observed object transfer; offline completion is labelled rehearsal.

Find one failure and test a repair

Use a controlled change and evidence to improve a sequence.

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You show a failing prediction, one repair and a repeated check with its limits.

Show one prediction and result

Explain what your program did and what the evidence cannot prove.

Our prediction was ...: _____

We changed ...: _____

Our evidence was ...: _____

This does not prove ...: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: All three pairs demonstrate and state a specific limitation.